COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE21S2
DATE PERFORMED:	02/09/26	**DATE SUBMITTED:**	02/09/26
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ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

What procedures did you perform?
1. Finding an Element

```
None
for (int i = 0; i < n; i++)
{
    if (arr[i] == target)
    {
        return i;
    }
}
```

Worst case:O(n)
Best case:O(1)
Explanation:Best case is if the target is located at index 1, Worst case is if the target is at the end of the array which is n.
2. Printing All Elements

```
None
for (int i = 0; i < n; i++)
{
    cout << arr[i] << endl;
}
```

Worst case:O(n)
Best case:O(n)
Explanation:Because all the n number of elements in the array has to be printed there is no early end.
3. Nested Loops

```
None
for (int i = 0; i < n; i++)
```

```
{
    for (int j = 0; j < n; j++)
    {
        cout << i << " " << j << endl;
    }
}
```

Worst case:O(n^2)
Best case:O(n^2)
Explanation:The worst and best case is the same because all you need to do is print the i and j for n*n which is n^2 times.
4. Conditional Processing

None
```
if (n > 100)
{
    for (int i = 0; i < n; i++)
    {
        cout << arr[i];
    }
}
else
{
    cout << "Small input";
}
```

Worst case:O(n)
Best case:O(1)
Explanation:The best case is O(1) because it will end the loop if the output is lower or equal to 100. The worst case is O(n) because if the n value is greater than 100 it will print out the n number of elements in the array.
5. Fixed Inner Loop

None
```
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < 10; j++)
    {
        cout << i << j;
    }
```

```
}
```

Worst case:O(n)
Best case:O(n)
Explanation:Best case and worst case is the same because the inner loop is only printing for 10 times not n* n and since the loop can also end if n is smaller or bigger than 10 that's why both best and worst case is O(n).

6. Two Consecutive Loops

```
None
for (int i = 0; i < n; i++)
{
    cout << arr[i];
}

for (int j = 0; j < n; j++)
{
    cout << arr[j] * 2;
}
```

Worst case:O(n)
Best case:O(n)
Explanation: Because the expression is O= 2n from n + n as it only has to print the same array but in different loop and constant is not considered in evaluating time complexity
7. Dividing by Two

```
None
int x = n;

while (x > 1)
{
    x = x / 2;
}
```

Worst case:O(log n)
Best case:O(log n)
Explanation:The best and worst case is O(log n) because it has to divide x with base 2 until it's equal or less than to 1.
8. Nested Loop With Different Limits

None
```
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < i; j++)
    {
        cout << i << j;
    }
}
```

Worst case:O(n^2)
Best case:O(n^2)
Explanation:The worst and best case is O(n^2) because the inner loop (j) will continue until the outer loop(i) is equal or greater than n making the loop run for n^2 times.
9. Early Termination

None
```
for (int i = 0; i < n; i++)
{
    if (arr[i] == target)
    {
        cout << "Found";
        break;
    }
}
```

Worst case:O(n)
Best case:O(1)
Explanation:The best case is O(1) because the loop will end if the target is found at the first index the worst case is O(n) if the target is found at the last index.
10. Two-Level Search

None
```
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        if (arr[i] == target)
        {
            break;
        }
```

```
    }
}
```

Worst case:O(n^2)
Best case:O(1)
Explanation:The best case is O(1) because the loop will break if the target is found at index 1 and the worst case is O(n^2) because the loop will go through 2 for loops for n x n times which is n^2 if the target is at the last index.
11. Linear Loop With Logarithmic Inner Work

None
```
for (int i = 0; i < n; i++)
{
    int x = n;

    while (x > 1)
    {
        x = x / 2;
    }
}
```

Worst case:O(N log n)
Best case:O(N log n)
Explanation: The worst and best case is O(N log n) because the loop has to run for n times until x is less than or equal to 1.
12. Conditional Nested Loop

None
```
if (arr[0] == target)
{
    cout << "Found";
}
else
{
    for (int i = 0; i < n; i++)
    {
        for (int j = 0; j < n; j++)
        {
            cout << arr[i] + arr[j];
        }
```

```
    }
}
```

Worst case:O(n^2)
Best case:O(1)
Explanation:The best case is O(1) because if the target is the first index and worst case is O(n^2) because if the target is not found at the first index the loop will instead print the sum of the array at index i and j running for n*n times (n^2).
13. Finding the Maximum

None
```
int max = arr[0];

for (int i = 1; i < n; i++)
{
    if (arr[i] > max)
    {
        max = arr[i];
    }
}
```

Worst case: O(n)
Best case: O(n)
Explanation:The best case and worst case is the same because the loop only changes the value to max at i if the element at i is bigger than max.

14. Three Nested Loops

None
```
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        for (int k = 0; k < n; k++)
        {
            cout << i + j + k;
        }
    }
}
```

Worst case:O(n^3)
Best case:O(n^3)
Explanation:The best and worst case is O(n^3) because the loop will run for n*n*n times (n^3) printing the sum of i j and k.

15. Comprehensive Analysis

```
for (int i = 0; i < n; i++)
{
    if (arr[i] == target)
    {
        return i;
    }

    int x = n;

    while (x > 1)
    {
        x = x / 2;
    }
}
```

Worst case:O(N log n)
Best case:O(1)
Explanation:The best case is O(1) because the loop will end if the target was found at the first index and the worst case O(N log n) if the target is at the end of array the loop has to go through the array for n times and also has to divide x by 2 until x is equal to or less than 1.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or

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